

Intellectual Merit Criterion

Overall Assessment of Intellectual Merit

Excellent

Explanation to Applicant

The Candidate is a trained academic researcher - while still an undergraduate student at NCSU. He also has publication records and worked on several funded projects led by the faculty at NCSU. The support letters are all uniformly strong and laud the technical competence and work ethics of Kevin.

Broader Impacts Criterion

Overall Assessment of Broader Impacts

Very Good

Explanation to Applicant

The NSF GRF will offer the platform to realize Kevin's ambition, enabling him to pursue a doctorate focused on robotics, controls, and optimization. As renewable technology improves and as the scope for robotics grows, his doctorate education and research experience will enable him to continue to identify problems and derive optimal solutions for them.

Summary Comments

In all respects, the Candidate highly deserves the fellowship.

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Overall Assessment of Intellectual Merit

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Explanation to Applicant

The applicant has demonstrated high academic achievement, and has had involvement with dissemination of the products of their work through conference presentations. The applicant's personal statement demonstrates a strong enthusiasm and raw curiosity for their chosen field of interest, and the applicant has relevant experience in participating in research opportunities as well as taking advanced graduate courses as an undergraduate student. The research proposal clearly provides a testable hypothesis and a well-developed methodology; there are clearly identified metrics with which to evaluate the success of the project. The proposal is well-grounded in the literature and the project is of timely relevance to the applicant's chosen field of interest.

Broader Impacts Criterion

Overall Assessment of Broader Impacts

Very Good

Explanation to Applicant

The applicant identifies an awareness of students in their community facing barriers to STEM access, and they provide some examples of demonstrated service activities related to extracurricular projects, including the university rocket club. The proposed research is of benefit to society, but would benefit from additional specificity in the engagement of students in a research context. Nonetheless, the proposed effort is highly interdisciplinary, providing opportunities to identify and tackle the research problems with novel approaches.

Summary Comments

The applicant is highly accomplished and is clearly motivated to solve research problems related to robotics and controls, and this is holistically demonstrated across both the personal statement and the research proposal, and this is also corroborated by the reference letters. The applicant has near-term goals and milestones clearly outlined with respect to both their research career and the societal impact they want to have. Plans beyond graduate school are less clear, but this is not critical at this stage. The applicant has a demonstrated commitment to mentorship, and although systemic reasons for existing barriers to STEM are not identified with the specificity required to dismantle them, the applicant has high potential to positively impact marginalized populations.

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Explanation to Applicant

Strengths: Impressive academic performance in Aeronautical and Aerospace Engineering and Applied Mathematics at North Carolina State University. Extensive research experience, including significant contributions to the CORE Lab and the DARPA Manta Ray project, showcasing advanced technical skills and innovative problem-solving abilities. Proven ability to pursue and excel in advanced coursework and research, evident from early high school engagement in university-level math courses and later undergraduate research. Publication of research papers in prestigious conferences, demonstrating a high level of scientific communication and mastery in the field. Weaknesses: None.

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Explanation to Applicant

Strengths: Significant contributions to community service and STEM education through the InspireNC non-profit organization, reflecting a strong commitment to societal betterment and education. Leadership roles in various projects and teams, including the Liquid Rocketry Lab and InspireNC, demonstrating effective teamwork and project management skills. Potential for future broader impacts is high, with clear goals to enhance renewably powered robotics and their applications in diverse fields such as ocean monitoring and agriculture. Weaknesses: The broader impact on the academic community, while promising, could be more explicitly detailed in terms of planned outreach and collaboration beyond current achievements.

Summary Comments

The applicant exhibits exceptional intellectual merit and broader impacts, making them a highly competitive candidate for the NSF GRFP. The applicant's academic excellence is evident in their outstanding GPA and proactive engagement in advanced studies. Their research experience, particularly in the CORE Lab and DARPA Manta Ray project, highlights their ability to contribute significantly to advancing knowledge in aerospace engineering and controls. Additionally, their leadership in community outreach and STEM education through InspireNC underscores their commitment to societal improvement. The reference letters strongly support their capabilities and potential, underscoring their proactive approach and technical expertise. The applicant's future goals align well with the NSF's objectives, demonstrating a clear vision for furthering research in renewably powered robotics and its applications.

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Explanation to Applicant

The proposed research is an excellent advancement for the swarm of unmanned vehicles under imprecise information and communication. The stochastic differential game theory is not new, but the application is required in unmanned vehicles. The candidate has skills and background; his advisor is well placed to advise the candidate. The candidates aerospace fundamental classes grades are not 'A.' There is confusion between probability and statistics.

Broader Impacts Criterion

Overall Assessment of Broader Impacts

Excellent

Explanation to Applicant

Creation of information value of energy can be applied to other autonomous systems.

Summary Comments

The proposed research is well defined and required for the swarm of the unmanned vehicles. The candidate and his advisor are well placed to carry out it.